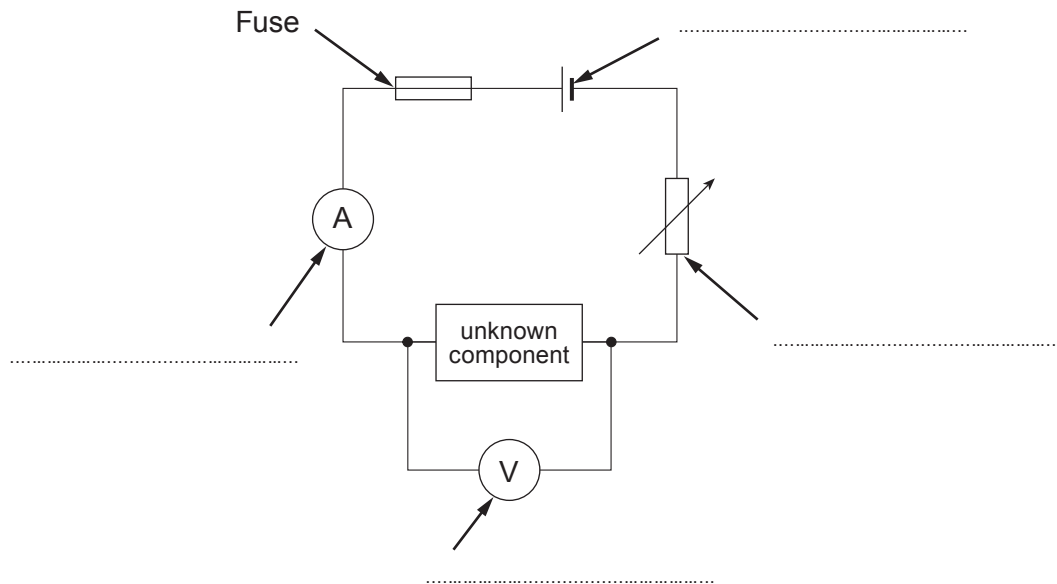


3. A student is given an unknown electrical component in a sealed box. He carries out an experiment to identify it.

He sets up the circuit below to investigate how the current changes with voltage for the unknown component.



- (a) **Complete the labelling** on the circuit diagram.

[4]

- (b) The data shown in the table is collected from the experiment.

Voltage (V)	-0.4	-0.2	0.0	0.2	0.4	0.6	0.8	1.0
Current (mA)	0	0	0	0	0	5	20	50

- (i) To prevent the unknown component being damaged the current through it **must not** be greater than 150 mA.

Circle the appropriate fuse that should be used in the circuit.

[1]

10 mA

100 mA

200 mA

500 mA

- (ii) (I) State the current in **amps (A)** when the voltage is 1.0 V.

[1]

Current = A

(II) Use the equation:

[2]

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

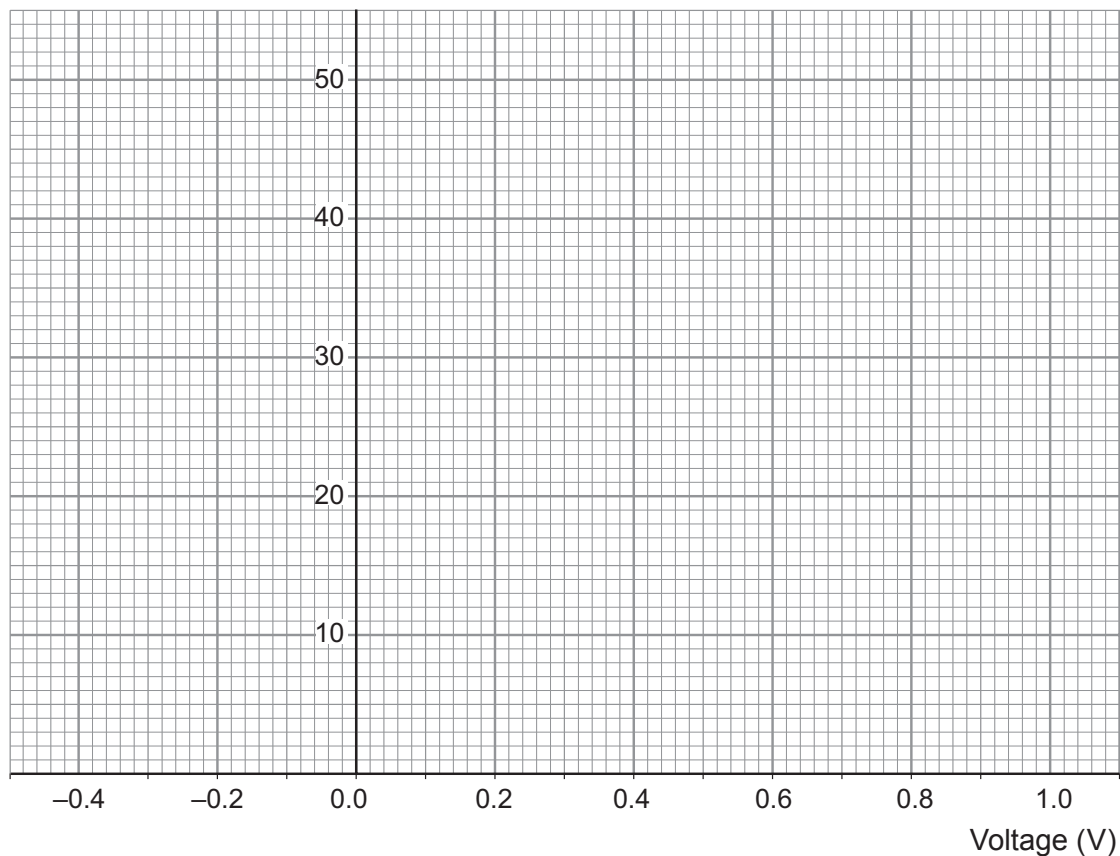
to calculate the resistance of the unknown component at 1.0 V.

Resistance = Ω

(iii) Plot the data in the table on the grid below and draw a suitable line.

[3]

Current (mA)



(iv) Use the graph to find the current when the voltage is 0.7 V.

[1]

Current = mA

(v) Describe the relationship between the current and the voltage in the range 0.4 V to 1.0 V.

[2]

.....

.....

.....

(vi) Identify the unknown component.

[1]